

Napkin-Style L^AT_EX Template: A Guide

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§1. Overview

This template is adapted from Evan Chen’s *An Infinitely Large Napkin* (<https://web.evanchen.cc/napkin.html>), a beautifully typeset open-access mathematics textbook. The template provides:

- Colored theorem, lemma, and example boxes using `tcolorbox`.
- Purple § section numbering in the KOMA-Script style.
- A rich collection of math macros for algebra, topology, and more.
- Sensible hyperlink colors and cross-referencing via `cleveref`.
- Support for commutative diagrams via `tikz-cd`.

§2. File Structure

The template consists of four files:

File	Purpose
<code>napkin-style.sty</code>	Visual style: layout, colors, theorem boxes
<code>napkin-macros.sty</code>	Math macros and operator definitions
<code>template.tex</code>	Minimal working document to start from
<code>guide.tex</code>	This documentation file

To start a new document, copy `template.tex` to your project folder. Both `.sty` files must be in the same folder as your `.tex` file, or installed to your local `texmf` tree (see [Section 3.ii](#)).

§3. How to Compile

The recommended way to compile any document using this template is with `latexmk`. Create a `.latexmkrc` file in your project directory:

```
push @extra_pdflatex_options, '-synctex=1';
$pdf_mode = 1;
$max_repeat = 5;
```

Then run:

```
latexmk -f yourfile.tex
```

`latexmk` automatically reruns `pdflatex` as many times as needed to resolve all cross-references and the table of contents.

§3.i. Manual compilation

If you prefer to call `pdflatex` directly, run it *twice*:

```
pdflatex yourfile.tex
pdflatex yourfile.tex
```

The first pass generates auxiliary files; the second pass reads them back to produce correct page numbers and hyperlinks.

§3.ii. Installing the style files globally

To use the style files from any directory without copying them each time, place them in your local `texmf` tree:

```
mkdir -p ~/texmf/tex/latex/napkin-style
cp napkin-style.sty napkin-macros.sty ~/texmf/tex/latex/napkin-style/
texhash ~/texmf
```

After this, you can `\usepackage{napkin-style}` from any project.

§4. Document Classes

The style works with both KOMA-Script classes.

Class	Best for	Header behavior
<code>scrartcl</code>	Problem sets, short notes, single-topic	Section title
<code>scrbook</code>	Multi-chapter notes, course notes	Chapter + section

The style file detects the class automatically. For a book-style document:

```
\documentclass[11pt,twoside=semi,openright]{scrbook}
\usepackage{napkin-style}
\usepackage{napkin-macros}
```

§5. Theorem Environments

All environments are numbered together within each section, so the counter resets at each new section. The available environments are listed below with live examples.

§5.i. Blue theorem box (theorem, lemma, proposition, corollary)

Theorem 5.1 (Lagrange)

If G is a finite group and $H \leq G$, then $|H|$ divides $|G|$.

Lemma 5.2

Every group of prime order is cyclic.

Proposition 5.3

The center $Z(G)$ is always a normal subgroup of G .

Corollary 5.4

If $|G| = p$ is prime, then $G \cong \mathbb{Z}/p\mathbb{Z}$.

§5.ii. Orange example box

Example 5.5

The symmetric group S_3 has order $6 = 3!$. Its center is trivial: $Z(S_3) = \{e\}$.

§5.iii. Green remark box

Remark 5.6 — Lagrange's theorem has no converse in general: a group of order 6 need not have a subgroup of every order dividing 6.

§5.iv. Question and exercise box

Question 5.7. Does every group of order 4 have a subgroup of order 2?

Exercise 5.8. Prove that any group of order p^2 (for prime p) is abelian.

§5.v. Definition and claim (plain style)

Definition 5.9. A **normal subgroup** $N \trianglelefteq G$ satisfies $gNg^{-1} = N$ for all $g \in G$.

Claim 5.10. The kernel of any homomorphism is normal.

§5.vi. Subproof

Inside a proof, you can use `subproof` to delimit a sub-argument with a filled square ■ as the QED symbol:

Proof. We show two things.

Step 1. The kernel is a subgroup (closure under product and inverse). ■

Step 2. It is closed under conjugation because homomorphisms respect products. ■

Both steps together complete the proof. □

§6. Special Commands

§6.i. `\prototype`

Use `\prototype{...}` at the start of a section to highlight the canonical motivating example. It renders in red italic.

Prototypical example for this section: The integers $\mathbb{Z}$ under addition.

The text of the section then continues normally after the prototype line.

§6.ii. `\moral environment`

The `\moral` environment draws a green bordered box for the key conceptual takeaway of a section.

Normal subgroups are exactly the subgroups that arise as kernels of homomorphisms.

§6.iii. `\vocab`

`\vocab{term}` typesets a vocabulary term in **blue bold**, useful when introducing a new definition inline.

§6.iv. `\hrulebar`

`\hrulebar` inserts a thin horizontal rule, useful as a visual separator.

Text continues here after the rule.

§7. Commutative Diagrams

The `tikz-cd` package is loaded and configured with hook arrows matching Napkin's style.

Example 7.1

The first isomorphism theorem gives the commutative diagram

$$\begin{array}{ccc} G & \xrightarrow{\phi} & H \\ \pi \downarrow & \searrow \bar{\phi} & \\ G/\ker \phi & & \end{array}$$

where π is the quotient map and $\bar{\phi}$ is the induced injection.

Useful `tikz-cd` arrow options:

<code>two heads</code>	surjective $\twoheadrightarrow$
<code>hook</code>	injective $\hookrightarrow$
<code>dashed</code>	dashed arrow
<code>squiggly</code>	squiggly arrow
<code>Rightarrow</code>	double arrow $\Rightarrow$

§8. Math Macros Reference

§8.i. Number fields

Command	Output	Meaning
<code>\NN</code>	$\mathbb{N}$	Natural numbers
<code>\ZZ</code>	$\mathbb{Z}$	Integers
<code>\QQ</code>	$\mathbb{Q}$	Rationals
<code>\RR</code>	$\mathbb{R}$	Reals
<code>\CC</code>	$\mathbb{C}$	Complex numbers
<code>\FF</code>	$\mathbb{F}$	Generic field
<code>\EE</code>	$\mathbb{E}$	Expectation / generic field

§8.ii. Common operators

<code>\abs{x}</code>	$ x $	Absolute value
<code>\norm{x}</code>	$\ x\ $	Norm
<code>\floor{x}</code>	$\lfloor x \rfloor$	Floor
<code>\ceiling{x}</code>	$\lceil x \rceil$	Ceiling
<code>\cbrt{x}</code>	$\sqrt[3]{x}$	Cube root
<code>\eps</code>	ε	ε
<code>\defeq</code>	$:=$	Definition equality
<code>\injto</code>	$\hookrightarrow$	Injection arrow
<code>\surjto</code>	$\twoheadrightarrow$	Surjection arrow

§8.iii. Algebra shorthands

<code>\Zc{n}</code>	$\mathbb{Z}/n\mathbb{Z}$	$\mathbb{Z}/n\mathbb{Z}$
<code>\Zm{n}</code>	$(\mathbb{Z}/n\mathbb{Z})^\times$	$(\mathbb{Z}/n\mathbb{Z})^\times$
<code>\normalin</code>	$\trianglelefteq$	Normal subgroup
<code>\GL, \SL, \Gal, \Aut, \Hom, \End</code>	Standard algebra operators	

§8.iv. Category theory

<code>\catname{Grp}</code>	Grp	Category name (sans-serif)
<code>\op</code>	$\mathop{\cdot}^{\mathrm{op}}$	Opposite category
<code>\obj</code>	obj	Object class
<code>\Hom</code>	Hom	Hom-set

§8.v. Script letters

`\SA, \SB, \SC, \SD, \SF, \SG, \SH` produce $\mathcal{A}, \mathcal{B}, \mathcal{C}, \mathcal{D}, \mathcal{F}, \mathcal{G}, \mathcal{H}$ (`\mathscr`).
`\AA, \BB, \OO, \PP, \MM` produce $\mathcal{A}, \mathcal{B}, \mathcal{O}, \mathcal{P}, \mathcal{M}$ (`\mathcal`).

§8.vi. Cross-referencing with `cleveref`

Instead of `\ref`, use `\cref` and `\Cref`:

<code>\cref{thm:lagrange}</code>	“theorem 3.1” (lowercase)
<code>\Cref{thm:lagrange}</code>	“Theorem 3.1” (capitalized)

`cleveref` automatically inserts the correct environment name. Always label your environments:

```
\begin{theorem}
\label{thm:my-theorem}
...
\end{theorem}
```

§9. Customization

§9.i. Changing theorem colors

The box styles are defined in `napkin-style.sty` as `tcbset` keys. To change the theorem box color, add this after `\usepackage{napkin-style}`:

```
\tcbset{
  theorem box/.style={
    enhanced, arc=9pt, colframe=Plum, colback=Plum!5,
    boxrule=1pt, before skip=12pt, after skip=14pt,
    left=10pt, right=10pt
  }
}
```

§9.ii. Changing the § symbol

To remove the section symbol prefix, comment out or change these lines in `napkin-style.sty`:

```
\renewcommand*{\sectionformat}{\color{purple}\S\thesection\autodot\enskip}
\renewcommand*{\subsectionformat}{\color{purple}\S\thesubsection\autodot\enskip}
```

§9.iii. Adding a bibliography

To add references, load `biblatex` after both style packages:

```
\usepackage{napkin-style}
\usepackage{napkin-macros}
\usepackage[backend=biber, style=alphabetic]{biblatex}
\addbibresource{references.bib}
```

Then place `\printbibliography` at the end of the document, and compile with:

```
latexmk -f yourfile.tex % latexmk runs biber automatically
```